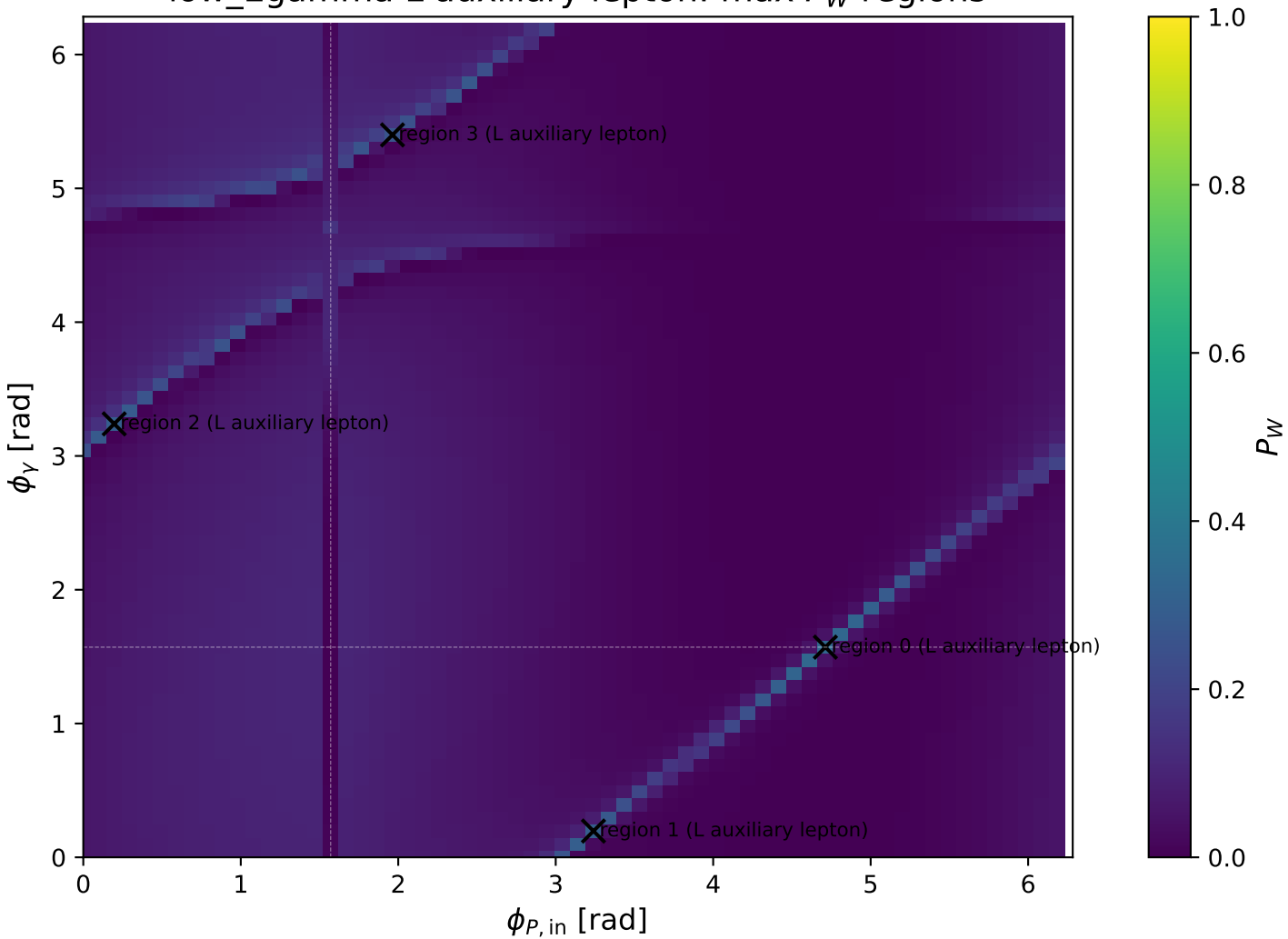
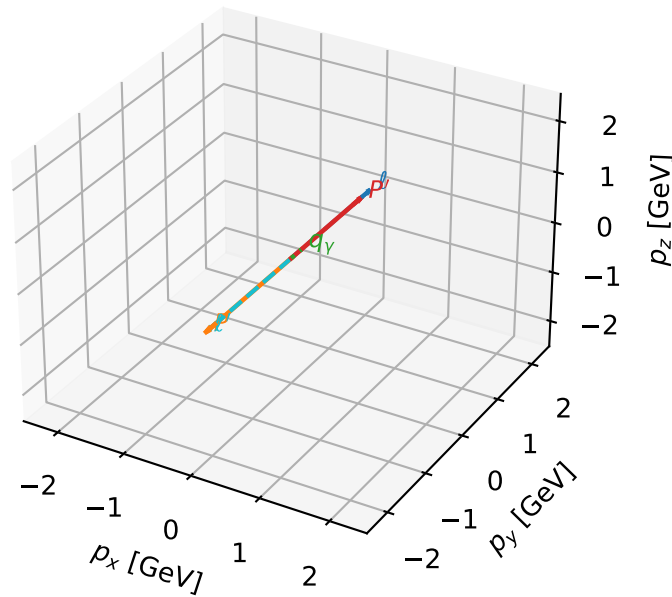


low_Egamma L auxiliary lepton: max P_W regions



L auxiliary lepton characteristic max P_W configuration

3D momenta

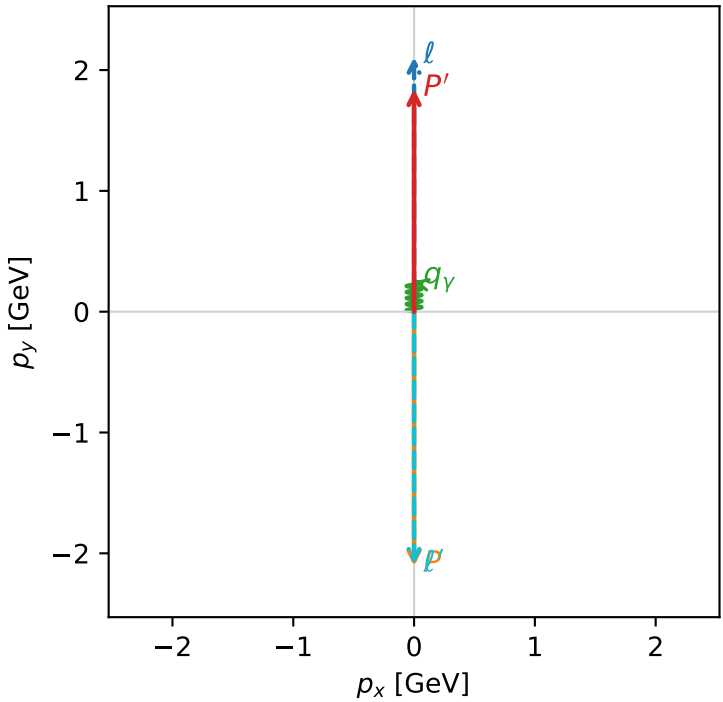


w_purity_low_egamma_l_lepton_region_0 P_W=0.333333
region: low_Egamma_L_lepton_region_0
outgoing-state purity: 0.994817
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.84352$
 $\theta_{in}=1.57079$, $\phi_\ell=1.5708$, $\phi_P=4.71239$
 $E_\gamma=0.25$, $\phi_\gamma=1.5708$
 $Q^2=17.6437$, $x_B=0.993667$, $t=-15.5495$, $W^2=0.9923$, $y=0.904266$
 $\theta(\ell', \gamma)=180$ deg

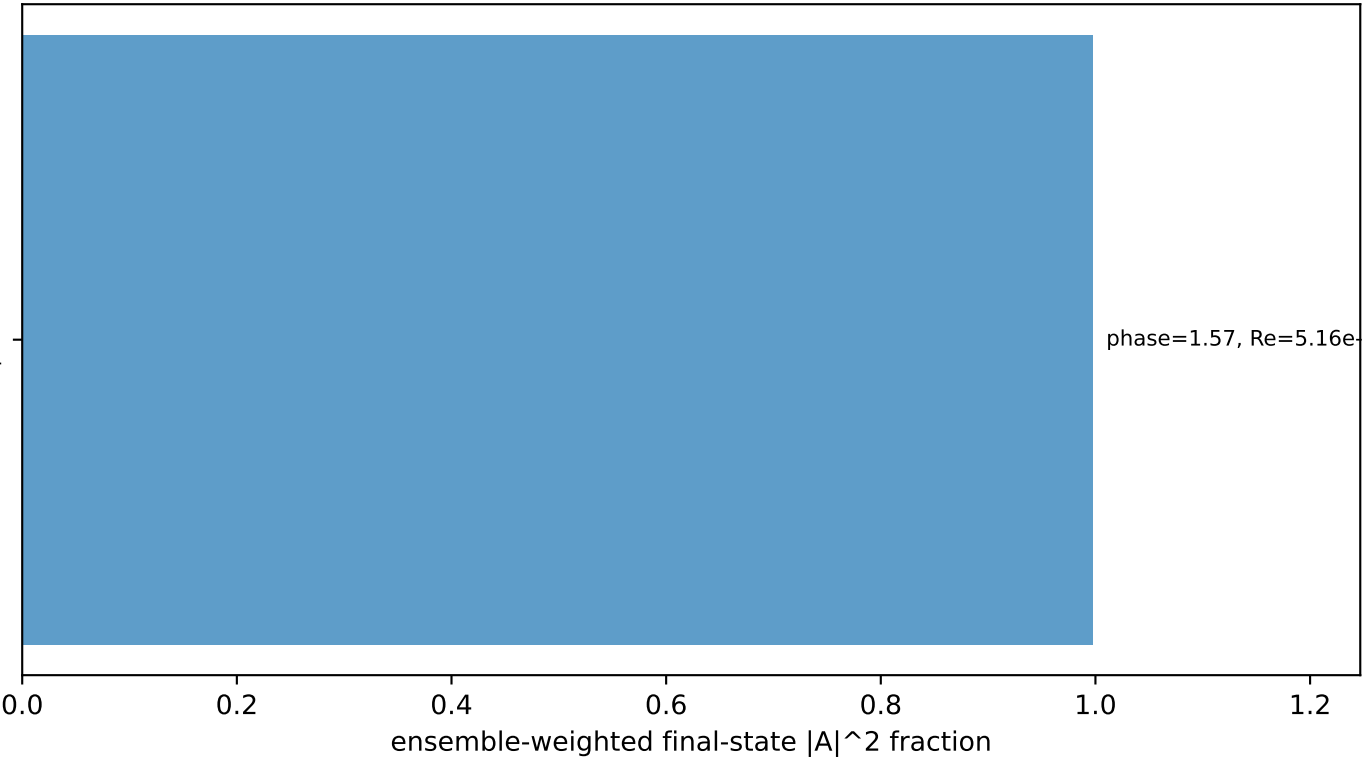
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.3201$ $p_x= -0.0000$ $p_y= -2.0935$ $p_z= 0.0000$
 P' $E= 2.0684$ $p_x= 0.0000$ $p_y= 1.8435$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
electron L, proton h=-1

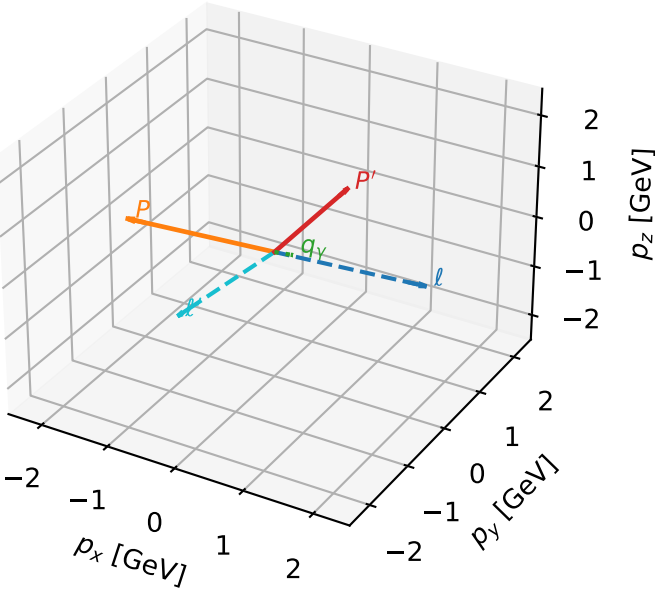
Leading initial-ensemble/final-state components (N=1, retained=99.7%)



phase=1.57, Re=5.16e-20, Im=1.41e-02

L auxiliary lepton characteristic max P_W configuration

3D momenta

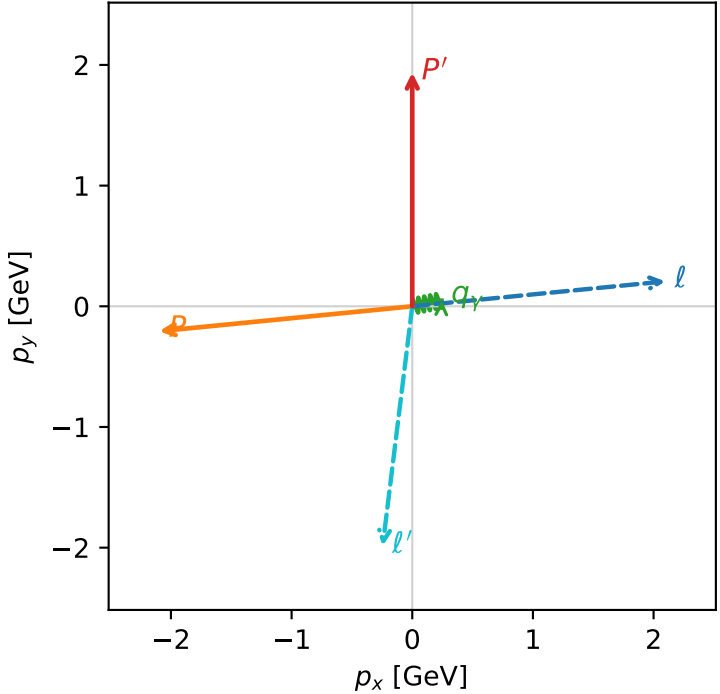


w_purity_low_egamma_l_lepton_region_1 $P_W=0.296822$
region: low_Egamma_L_lepton_region_1
outgoing-state purity: 0.643975
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_l, h_p)$

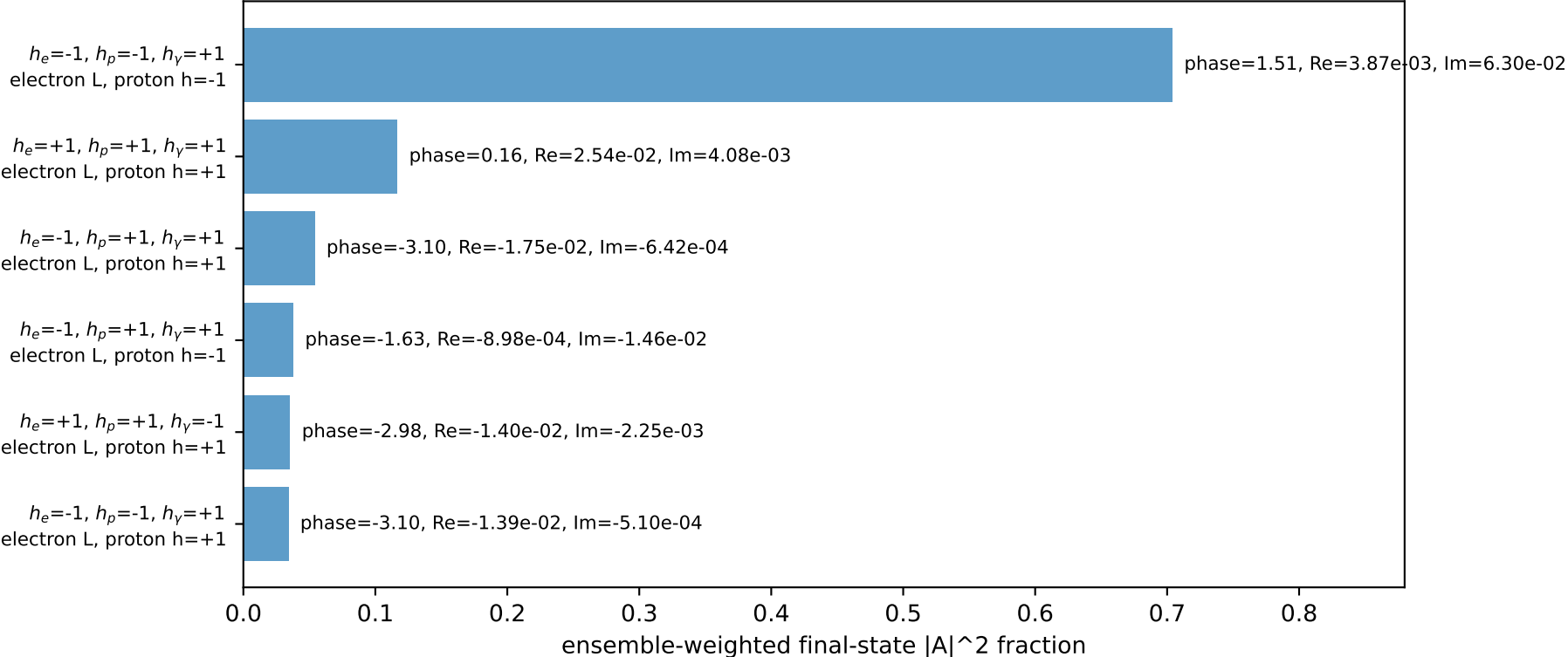
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.93673$
 $\theta_{in}=1.57079$, $\phi_l=0.0981748$, $\phi_P=3.23977$
 $E_\gamma=0.25$, $\phi_\gamma=0.19635$
 $Q^2=10.2807$, $x_B=0.920571$, $t=-8.96624$, $W^2=1.76689$, $y=0.568741$
 $\theta(l', \gamma)=108.29$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 2.0968$ $p_y= 0.2065$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -2.0968$ $p_y= -0.2065$ $p_z= 0.0000$
 l' $E= 2.2366$ $p_x= -0.2452$ $p_y= -1.9855$ $p_z= 0.0000$
 P' $E= 2.1519$ $p_x= 0.0000$ $p_y= 1.9367$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2452$ $p_y= 0.0488$ $p_z= 0.0000$

Transverse plane

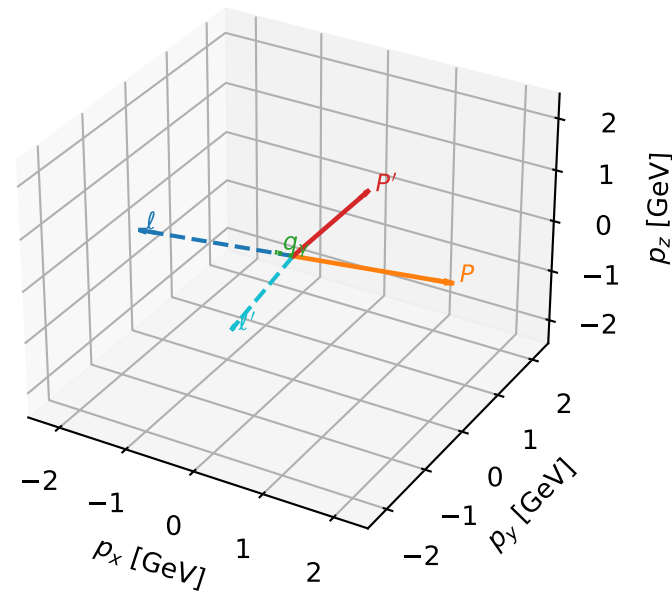


Leading initial-ensemble/final-state components (N=6, retained=98.2%)



L auxiliary lepton characteristic max P_W configuration

3D momenta

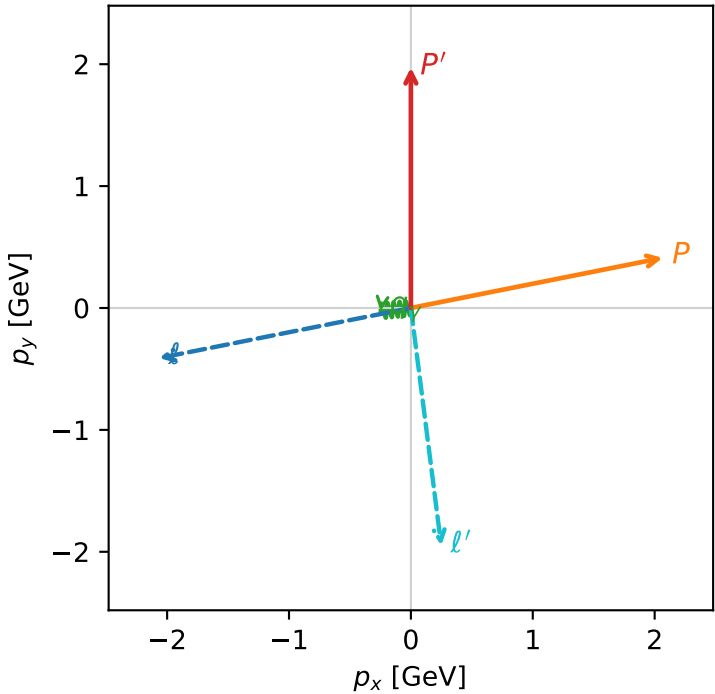


w_purity_low_egamma_l_lepton_region_2 P_W=0.293948
region: low_Egamma_L_lepton_region_2
outgoing-state purity: 0.674292
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_l, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.97282$
 $\theta_{in}=1.57079$, $\phi_l=3.33794$, $\phi_P=0.19635$
 $E_\gamma=0.25$, $\phi_\gamma=3.23977$
 $Q^2=7.7072$, $x_B=0.866355$, $t=-6.69456$, $W^2=2.06876$, $y=0.453051$
 $\theta(l', \gamma)=91.6522$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -2.0665$ $p_y= -0.4110$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 2.0665$ $p_y= 0.4110$ $p_z= 0.0000$
 l' $E= 2.2041$ $p_x= 0.2488$ $p_y= -1.9483$ $p_z= 0.0000$
 P' $E= 2.1845$ $p_x= 0.0000$ $p_y= 1.9728$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2488$ $p_y= -0.0245$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
electron L, proton h=-1

$h_e=-1, h_p=+1, h_\gamma=+1$
electron L, proton h=+1

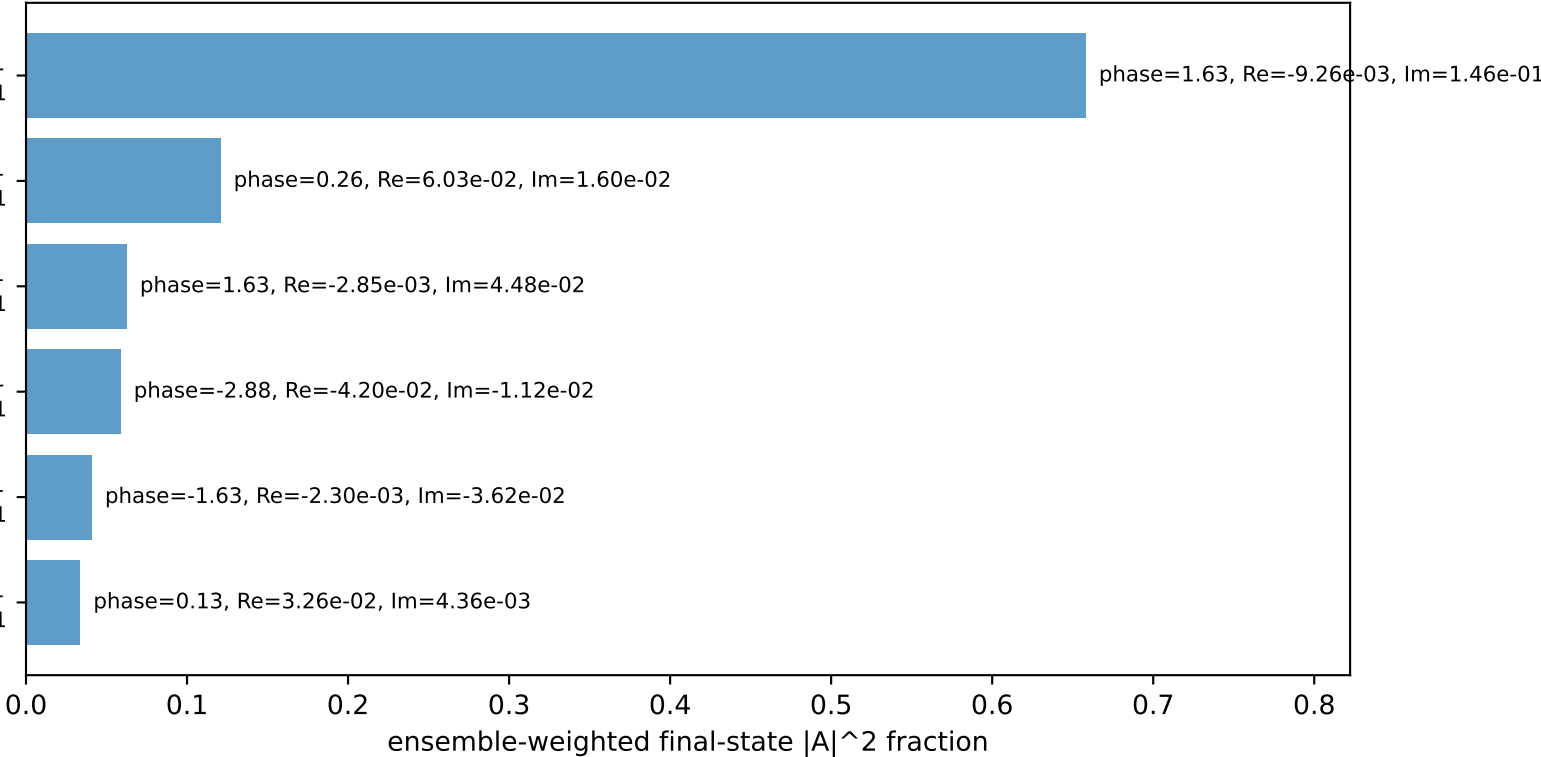
$h_e=-1, h_p=+1, h_\gamma=+1$
electron L, proton h=-1

$h_e=-1, h_p=-1, h_\gamma=+1$
electron L, proton h=+1

$h_e=+1, h_p=-1, h_\gamma=+1$
electron L, proton h=-1

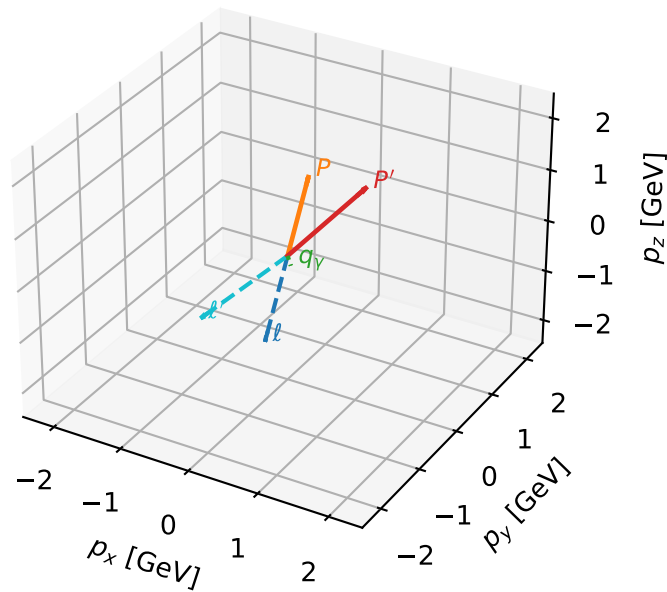
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton h=+1

Leading initial-ensemble/final-state components (N=6, retained=97.3%)



L auxiliary lepton characteristic max P_W configuration

3D momenta

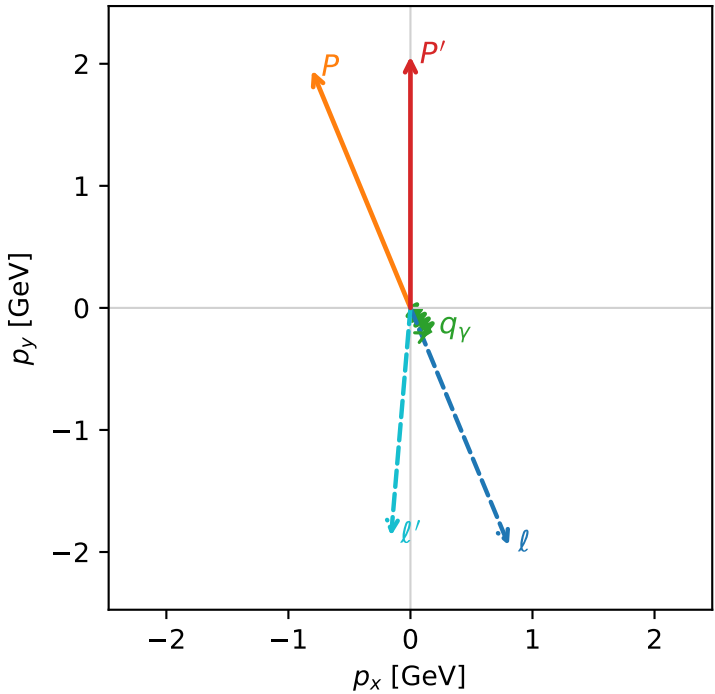


w_purity_low_egamma_l_lepton_region_3 P_W=0.262599
region: low_Egamma_L_lepton_region_3
outgoing-state purity: 0.510859
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_l, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.06076$
 $\theta_{in}=1.57079$, $\phi_l=5.10509$, $\phi_p=1.9635$
 $E_Y=0.25$, $\phi_Y=5.39961$
 $Q^2=0.894045$, $x_B=0.316741$, $t=-0.661375$, $W^2=2.80844$, $y=0.143748$
 $\theta(l', \gamma)=44.2292$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 0.8063$ $p_y= -1.9466$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.8063$ $p_y= 1.9466$ $p_z= 0.0000$
 l' $E= 2.1243$ $p_x= -0.1586$ $p_y= -1.8675$ $p_z= 0.0000$
 P' $E= 2.2642$ $p_x= 0.0000$ $p_y= 2.0608$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.1586$ $p_y= -0.1933$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=98.4%)

